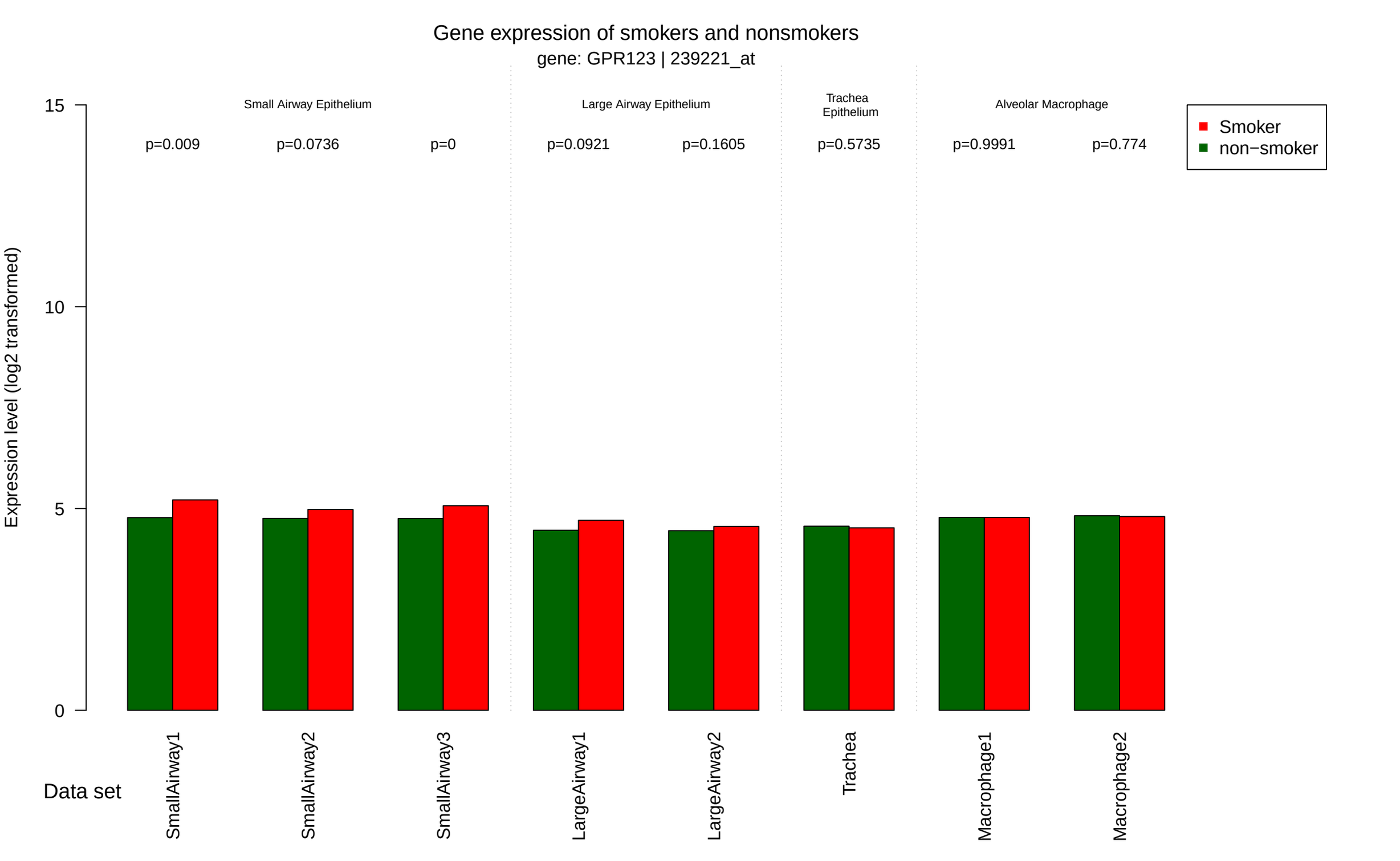
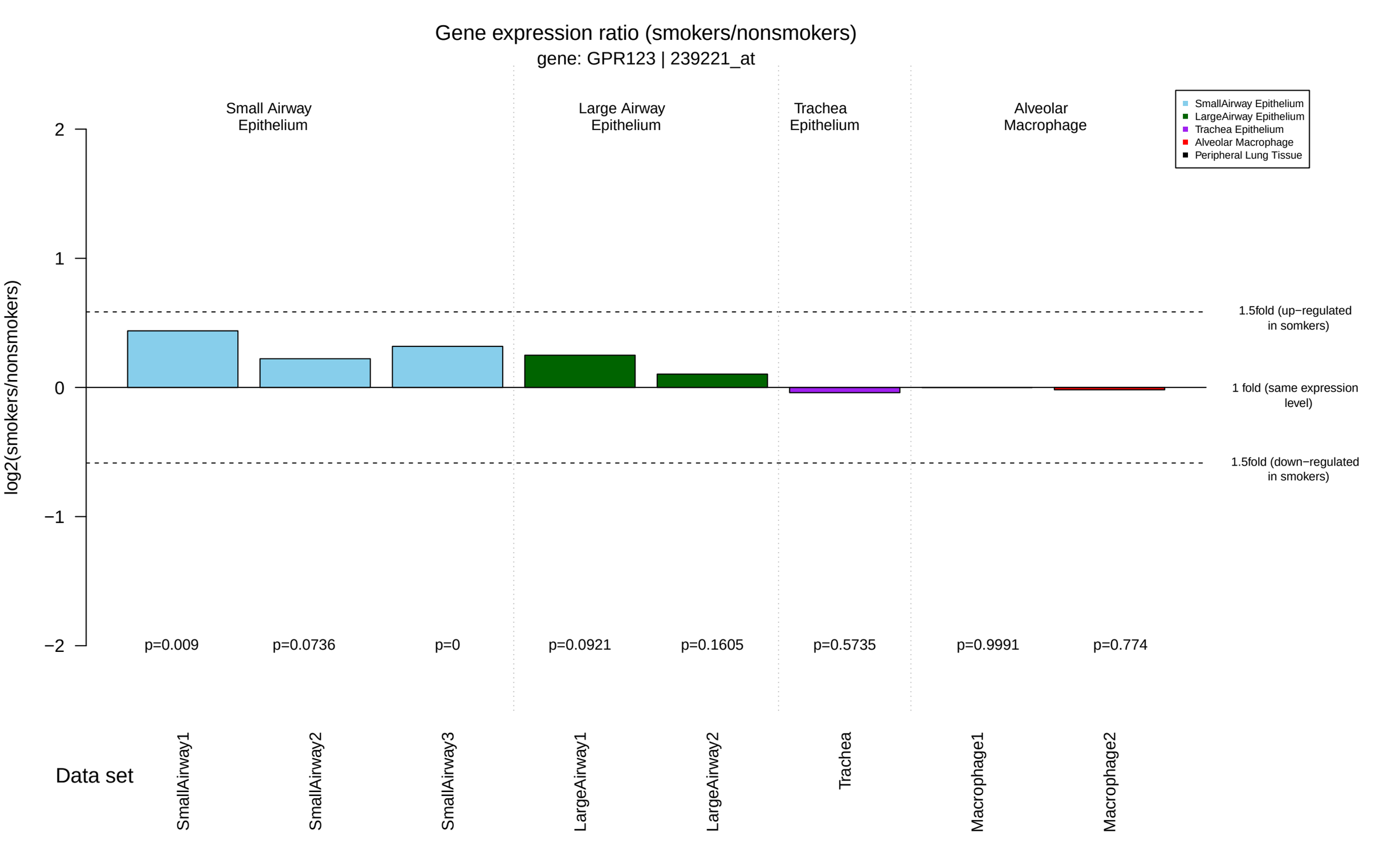
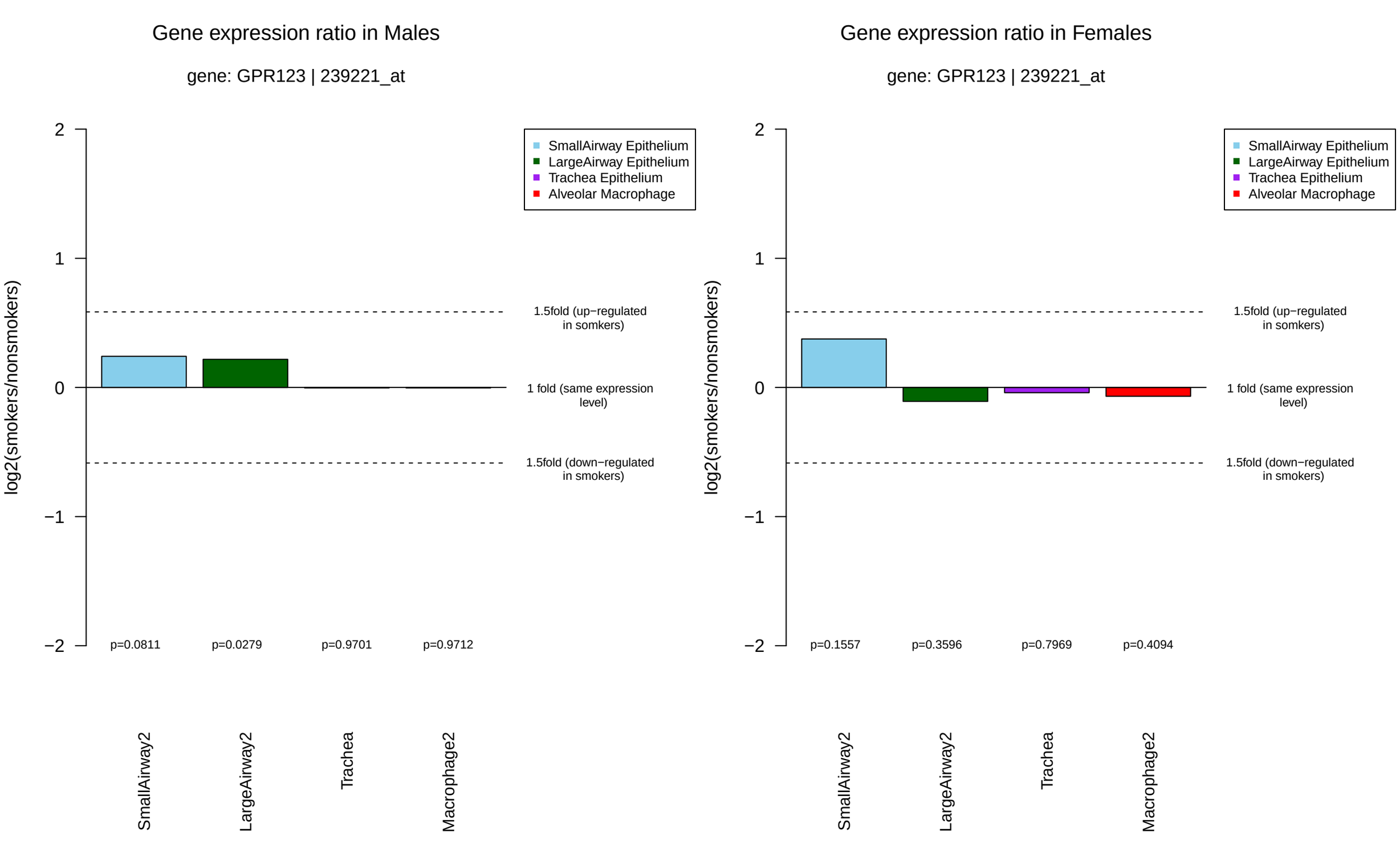
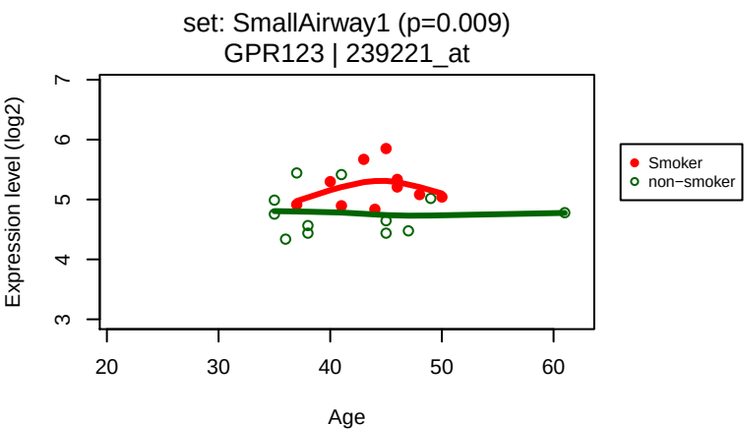
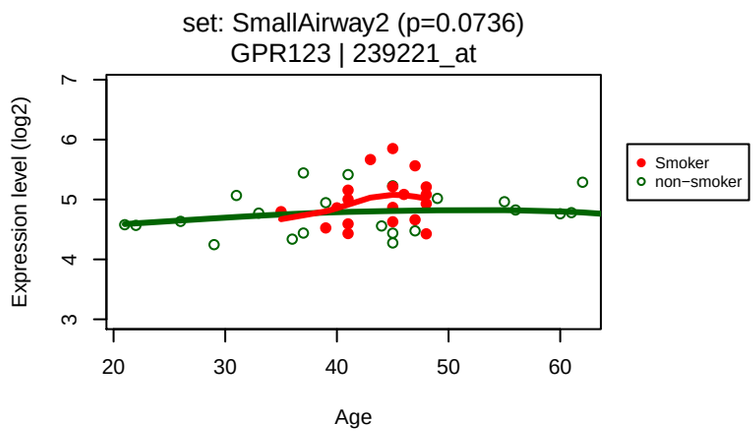
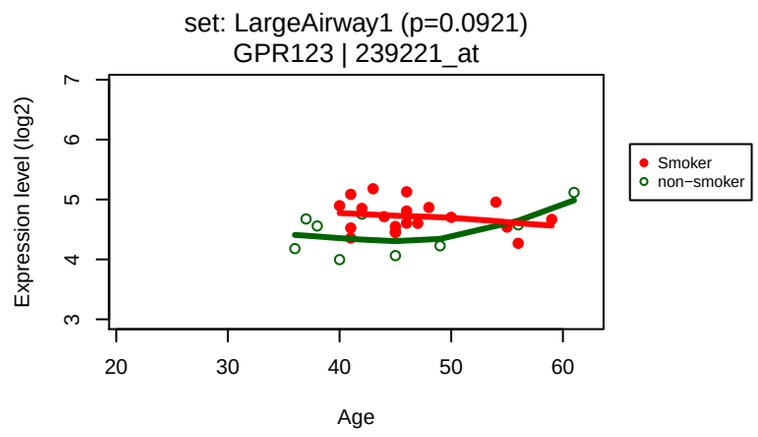

Gene expression of smokers and nonsmokers
gene: GPR123 | 239221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.009
p=0.0736
p=0
p=0.0921
p=0.1605
p=0.5735
p=0.9991
p=0.774
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

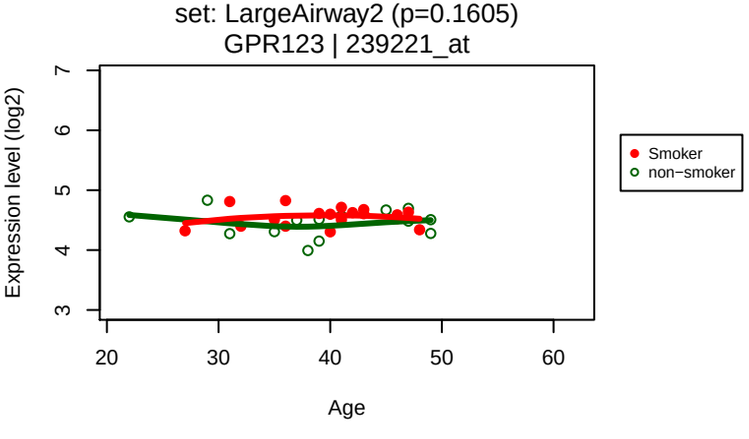

Gene expression ratio (smokers/nonsmokers)
gene: GPR123 | 239221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.009
p=0.0736
p=0
p=0.0921
p=0.1605
p=0.5735
p=0.9991
p=0.774
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

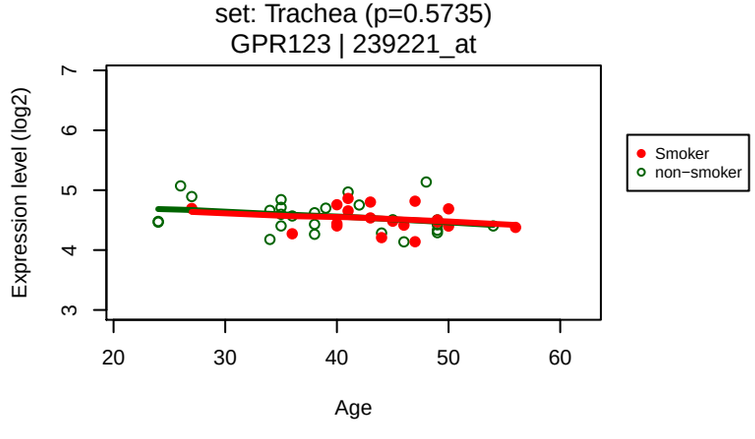

Gene expression ratio in Males
gene: GPR123 | 239221_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0811
p=0.0279
p=0.9701
p=0.9712
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR123 | 239221_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1557
p=0.3596
p=0.7969
p=0.4094
SmallAirway2
LargeAirway2
Trachea
Macrophage2

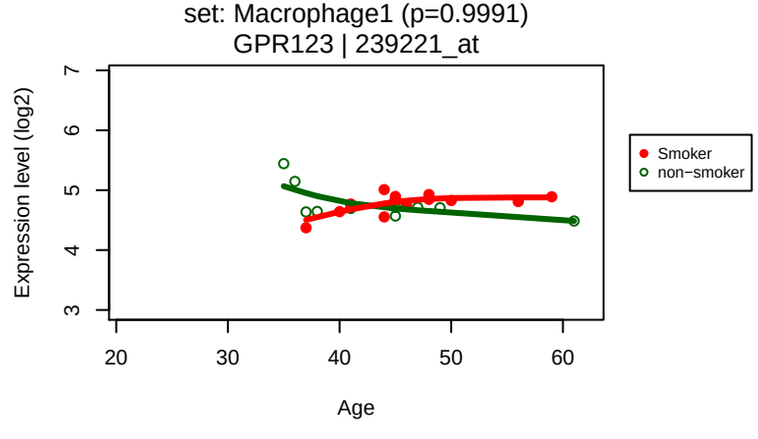

set: SmallAirway1 (p=0.009)
GPR123 | 239221_at
Expression level (log2)
Age
Smoker
non−smoker

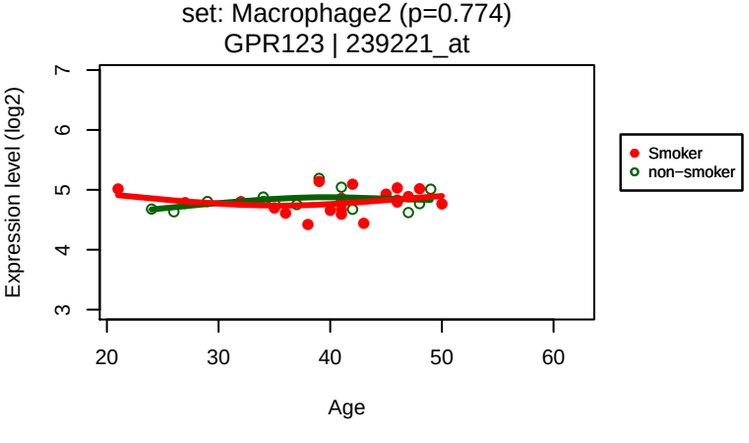

set: SmallAirway2 (p=0.0736)
GPR123 | 239221_at
Expression level (log2)
Age
Smoker
non−smoker

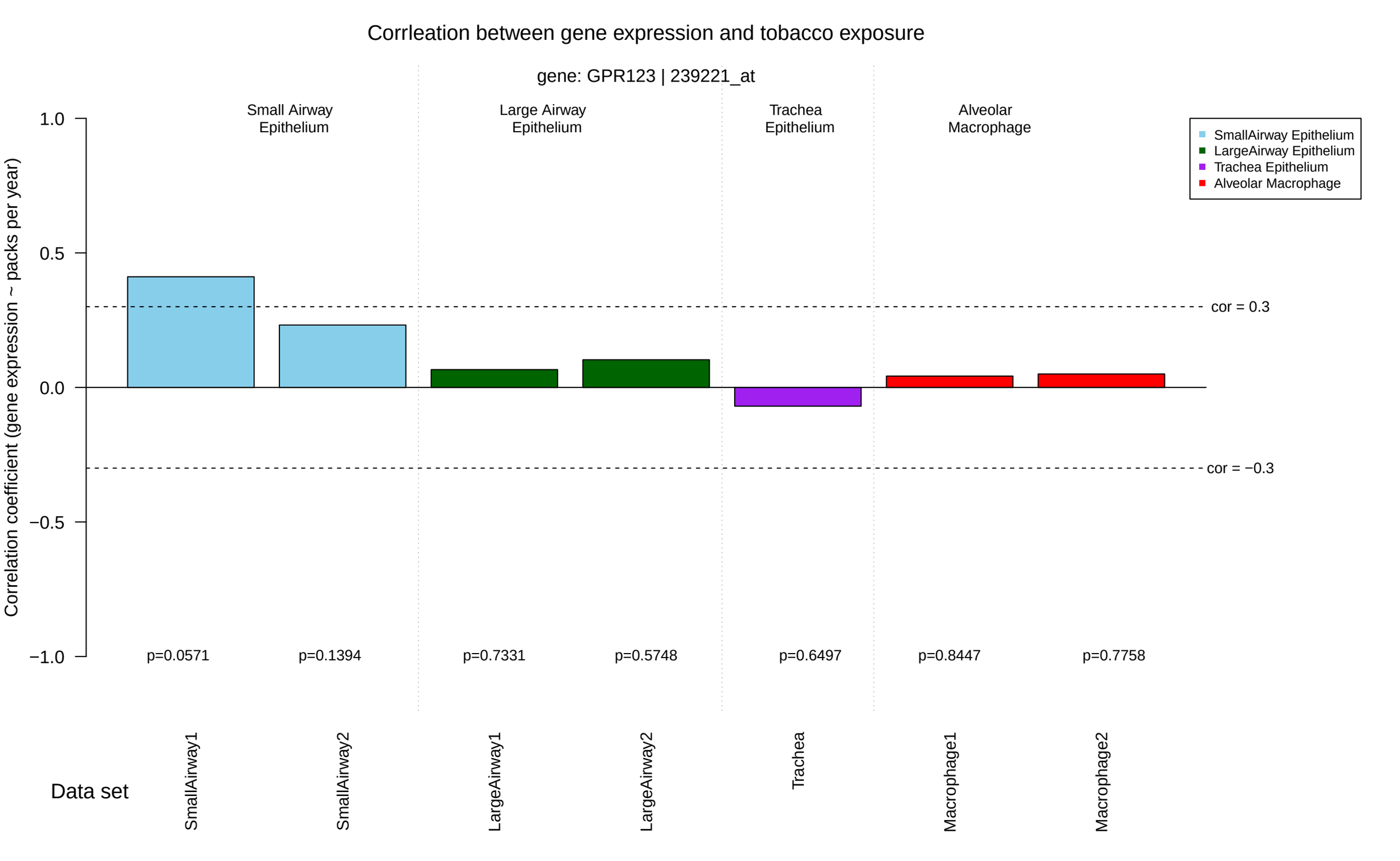

set: LargeAirway1 (p=0.0921)
GPR123 | 239221_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1605)
GPR123 | 239221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5735)
GPR123 | 239221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9991)
GPR123 | 239221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.774)
GPR123 | 239221_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR123 | 239221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0571
p=0.1394
p=0.7331
p=0.5748
p=0.6497
p=0.8447
p=0.7758
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2